NE 533 exam 2 Crystal B.

1)

$R_F = 0.55$ cm   LHR = 300 W/cm

(a) max stress of pellet?

$K_f = 0.025 \frac{W}{cm \cdot K}$ E = 200 GPa

$\nu = 0.35$ $\alpha = 10 \times 10^{-6}$ K^{-1}

σ_{hoop} = max stress
Fuel

$$\sigma_\theta (\eta) = -\sigma^* (1 - 3\eta^2)$$

$$\eta = \frac{r}{R_F^2}$$

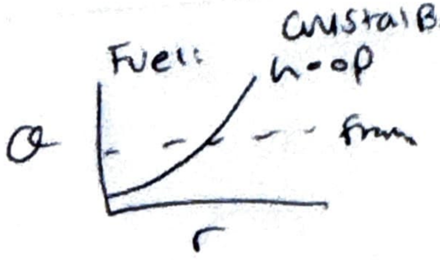
~~max~~ $T_0 - T_s = \frac{LHR}{4\pi K_f} = \frac{300}{4\pi \cdot 0.025}$

$= 954.9K$

$\sigma^* = \frac{\alpha E (T_0 - T_s)}{4(1-\nu)}$

$$\sigma^* = \frac{(10 \times 10^{-6} K^{-1})(200 GPa)(954.9K)}{4(1 - 0.35)} = 0.735$$

GPa = 735 MPa



1a)

$$\sigma_\theta = -735\text{ MPa}\,(1-3\eta^2) = 1470\text{ MPa}$$

$$\eta = \frac{r}{R_f} = \frac{R_f}{R_f} = 1$$

1b)

$\sigma_f = 175\text{ MPa}$

how far crack

$$\sigma_f = 175\text{ MPa} = -735\text{ MPa}\,(1-3\eta^2)$$

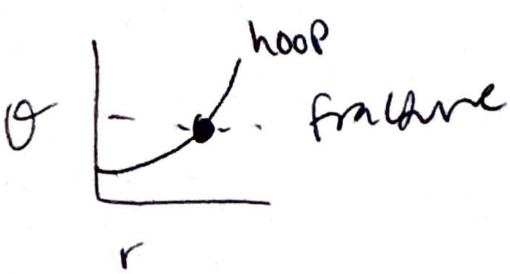
solve for η

$$-0.238 = 1-3\eta^2$$

$$-1.238 = -3\eta^2$$

$$\sqrt{0.413} = \sqrt{\eta^2}$$

$$\eta = 0.642$$



$$1-0.642 = 0.3576$$

35.8% of the way into the fuel is how far crack extends.

Crystal B.

2)

Clad

$P = 15\ \text{MPa}$ $\quad\quad \bar{R}_c = 0.55\ \text{cm}$

$t_c = 0.05\ \text{cm}$

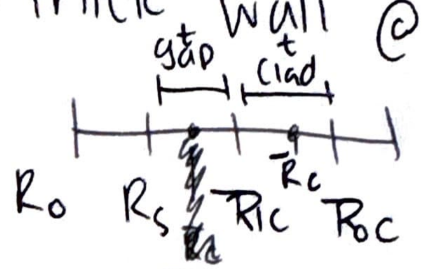
(a) thin wall:

$$\sigma_\theta = \frac{PR}{\delta} = \frac{(15\text{MPa})(0.55\cancel{\text{cm}})}{0.05\ \cancel{\text{cm}}} = 165\ \text{MPa}$$

$$\sigma_z = \frac{PR}{2\delta} = \frac{(15\text{MPa})(0.55\text{cm})}{2(0.05\text{cm})} = 82.5\ \text{MPa}$$

$$\sigma_r = -\frac{P}{2} = \frac{-15\ \text{MPa}}{2} = -7.5\ \text{MPa}$$

(b) Thick wall @ inside wall so R_{IC}



$\bar{R}_c = 0.55\ \text{cm}$

$t_c = 0.05\ \text{cm}$

$$\bar{R}_c = \frac{R_{IC} + R_{OC}}{2}$$

$$R_{IC} + \frac{t}{2} = \bar{R}_c$$

$$R_{OC} - \frac{t}{2} = \bar{R}_c$$

2b(cont.) Crystal B.

$$\bar{R}_c - \frac{t}{2} = R_{ic}$$

$$R_{ic} = 0.55\,cm - \frac{0.05}{2} = 0.525\,cm$$

$$R_{oc} = \bar{R}_c + \frac{t}{2} = 0.575\,cm$$

$$r = R_{ic} = 0.525\,cm$$

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2b cont.)

Thick wall!

$\frac{R_o}{r} = \frac{0.575}{0.525} = 1.095$

$\frac{R_o}{R_i} = \frac{0.575}{0.525}$

$$\sigma_r = -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1} = -15\text{MPa} \frac{(1.095)^2 - 1}{(1.095)^2 - 1}$$

$\sigma_r = -15\text{MPa}$

$$\sigma_\theta = p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} = 15\text{MPa} \frac{(1.095)^2 + 1}{(1.095)^2 - 1}$$

$\sigma_\theta = 165.735\text{MPa}$

$$\sigma_z = p \frac{1}{(R_o/R_i)^2 - 1} = 15\text{MPa} \left(\frac{1}{(1.095)^2 - 1}\right)$$

$\sigma_z = 75.37$ MPa

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3) Calc Δ Find Δt_{gap}

First

1) $T_0 - T_s$, $T_s - T_{IC}$, $T_{IC} - T_{OC}$

2) Find T_0, T_s, T_{IC}

given $T_{OC} = 550K$

$$T_0 - T_s = \frac{LHR}{4\pi K_f} = \frac{250 \text{ W/cm}}{4\pi (0.03)} = 663.1\text{ K}$$

$$T_s - T_{IC} = \frac{LHR}{2\pi R_f} \frac{t_g}{K_g} = \frac{250}{2\pi (0.48)} \frac{0.008 \text{ cm}}{0.003}$$

$$T_s - T_{IC} = 221\text{ K}$$

$$T_{IC} - T_{OC} = \frac{LHR}{2\pi R_f} \frac{t_c}{K_c} = \frac{250}{2\pi (0.48)} \frac{0.08}{0.15}$$

$$= 44.2\text{ K}$$

3 (cont.)

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$T_{IC} - T_{OC} = T_{IC} - 550K = 44.2K$

$T_{IC} = 594.2K$

$T_S - T_{IC} = T_S - 594.2K = 221K$

$T_S = 815.2K$

$T_O - T_S = 663.1$

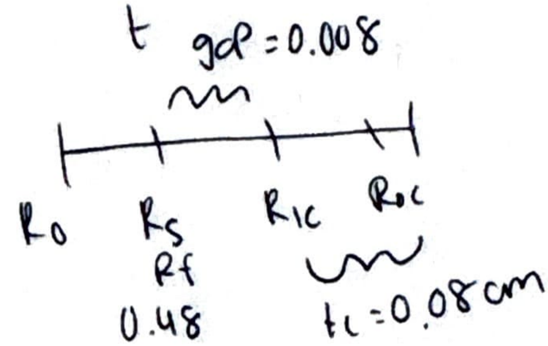
$T_O - 815.2 = 663.1$

$T_O = 1478.3K$

$T_{ref} = 300K$

$$\Delta S_{gap} = \bar{R}_c \alpha_c (\bar{T}_c - T_{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref})$$

$$\bar{R}_c = \frac{R_{IC} + R_{OC}}{2}$$



$$\bar{R}_c = 0.48\,cm + 0.008cm + \frac{0.08cm}{2}$$

$\bar{R}_c = 0.528\ cm$

$$\bar{T}_c = \frac{T_{IC} + T_{OC}}{2} = \frac{594.2 + 550K}{2} = 572.1K$$

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3 (cont.)

$$\Delta \delta_{gap} = \left[(0.528\,cm)(10\times10^{-6}\,K^{-1})(572.1K - 300K)\right] -$$

$$\left[(0.48\,cm)(14\times10^{-6}\,K^{-1})(1146.75\,K - 300K)\right]$$

$$\Delta \delta_{gap} = -0.00425\,cm$$

$$\overline{T}_f = \frac{T_0 + T_s}{2} = \frac{815.2 + 1478.3}{2} = 1146.75$$

$$[0.0014367] - [0.00569]$$

$$= -0.00425\ cm$$

exam 2

Crystal B

Cladding

4) hoop and rad stress $@_{r=R_{IC}}$

bc thermal expansion

$\Delta T = 50K$

$\alpha_c = 15 \times 10^{-6}$ $E = 100\ GPa$

$\nu = 0.34$

$t_c = 0.06\ cm$ $R_i = 0.55cm$

$r = R_{ic} = R_i = 0.55cm$

$$\sigma_r = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu}\left(\frac{r}{R_i}-1\right)\left(1-\frac{R_i}{\delta}\left(\frac{r}{R_i}-1\right)\right)$$

$$\sigma_\theta = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu}\left(1-2\frac{R_i}{\delta}\left(\frac{r}{R_i}-1\right)\right)$$

$$\sigma_r = \frac{1}{2}(50K)\frac{(15\times10^{-6})(100\ GPa)}{1-0.34}\left(\cancel{\frac{0.55}{0.55}-1}^{0}\right)\left(1-\frac{0.55}{0.06}\left(\frac{0.55}{0.55}-1\right)^{0}\right)$$

$\sigma_r = 0\ GPa$

Crystal Banes

4 cont.)

$$\sigma_\theta = \frac{1}{2}(50k)\frac{(15\times10^{-6})(100\,GPa)}{1-0.34}\left(1-2\frac{0.55}{0.06}\left(1-1^{0}\right)\right)$$

$$\sigma_\theta = 0.0568\,GPa$$

56.8 MPa

NE 532 Exam 2

Crystal Bowers

5) densification

$$\varepsilon_D = \Delta\rho_0 \left(\exp\left[\frac{\beta \ln 0.01}{C_D B_D}\right] - 1 \right)$$

$$\beta(FIMA) = \frac{\dot{F} t}{N_U} = \frac{(6\times10^{13})(864000\ sec)}{2.45\times10^{22}}$$

$\beta = 0.00212$ FIMA

$$N_U = \frac{\rho N_a}{A} = \frac{(10.97\ g/cm^3)(6.022\times10^{23}\ atom/mol)}{270\ g/mol} \frac{1U}{1UO_2} = 2.45\times10^{22}\ atom/cm^3$$

$$t = 10\ day\left(\frac{24\ h}{1\ day}\right)\left(\frac{60\ min}{1\ h}\right)\left(\frac{60\ sec}{1\ min}\right) = 864000\ s$$

$$\varepsilon_D = 0.015\left(\exp\left[\frac{(0.00212\ FIMA)\ln 0.01}{(1)(0.00526\ FIMA}\right] - 1\right.$$

$\varepsilon_D = -0.0127$ FINAL ANSWER

$C_D = 1$ $T > 750$ assume

$$B_D = \frac{5\ MWd/U}{950} = 0.00526\ FIMA$$

NES 33 Exam 2 Crystal Balls

$K_B = 8.1673 \times 10^{-5}$ eV/K
1 eV = 1.602×10^{-19} J
B_D = 5 MWD/kg
Start: 11:03 am
end: 12:18 pm
11:23 am

6) Elasticity is the ability of a mat'l to resist distortion and return to its original shape/size once the load is removed.

Plasticity is the ability of a material to plastically deform; non reversible change in size once load is applied.

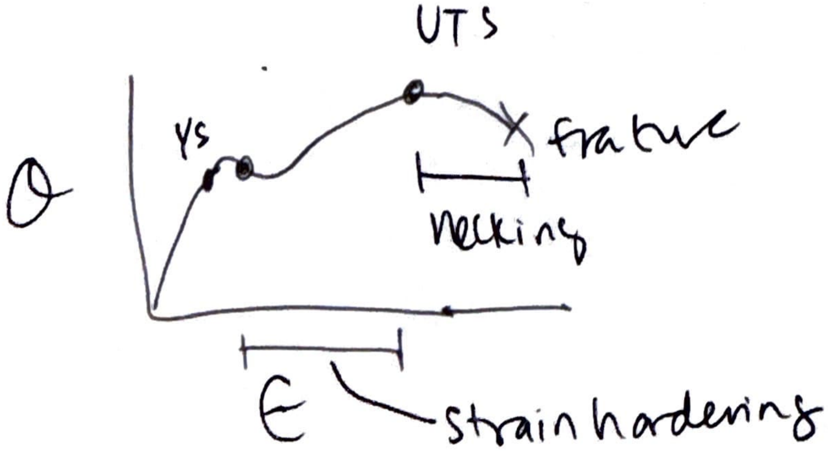
Crystal B.

7)

Strain hardening occurs after yielding in which a material hardens and increases in strength due to dislocation pileup.

Barriers cause dislocation mobility to decrease, and lead to pileup. A Barrier example is grain boundaries.

After the ultimate tensile strength, the material starts to neck.



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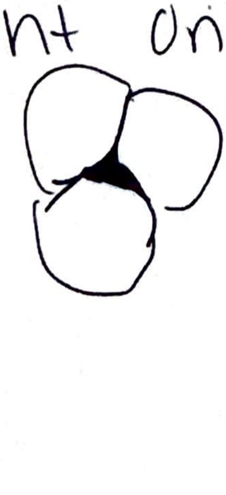
8) Fuel performance code must be able to predict fuel temperatures, specifically To centerline, predict cladding stress, and consider gap heat transfer, pressure, and gap closure.

9) Grains form in sintering Process as powder is compacted at high temp. / pressure w/o melting. Particles begin contacting each other at different orientations to form grains.



Crystal B.

10) mechanistic fuel performance modeling takes into account microstructure considerations and connects it to properties. mechanistic model relies on equations/physics rather than experimental data. It's beneficial because unlike empirical models, you can reach conditions outside of what the experimental data covers and make meaningful predictions and it performs comparable and sometimes better than empirical models.

crystal

11) microstructure is what can be observed at a magnification level of 25x like grains. A post-processing technique is working such as hot rolling in which metal is plastically deformed above the recrystallization temperature so new grains can be formed, which is the impact on microstructure.

Related properties impact is increasing the yield strength.

~~[illegible]~~ Crystal B.

12) Diffusion occurs more quickly on the grain boundaries because there is more open space.

13) Driving force for grain growth is reducing ~~[illegible]~~ grain boundary energy. Barriers like precipitates/ solute atoms and other defects can impede grain growth.

14) Fuel densifies during operation because fuel pellets are ~95% dense so in the reactor, the sintering process continues and small pores close (because of vacancy diffusion). ~~[illegible]~~ Driving ~~[illegible]~~ force is reduction in surface areas of pores.